Dynamic Motion: 20 s Vertical Drop & Gravity Support (2 kg Object Released at t = 1 s)

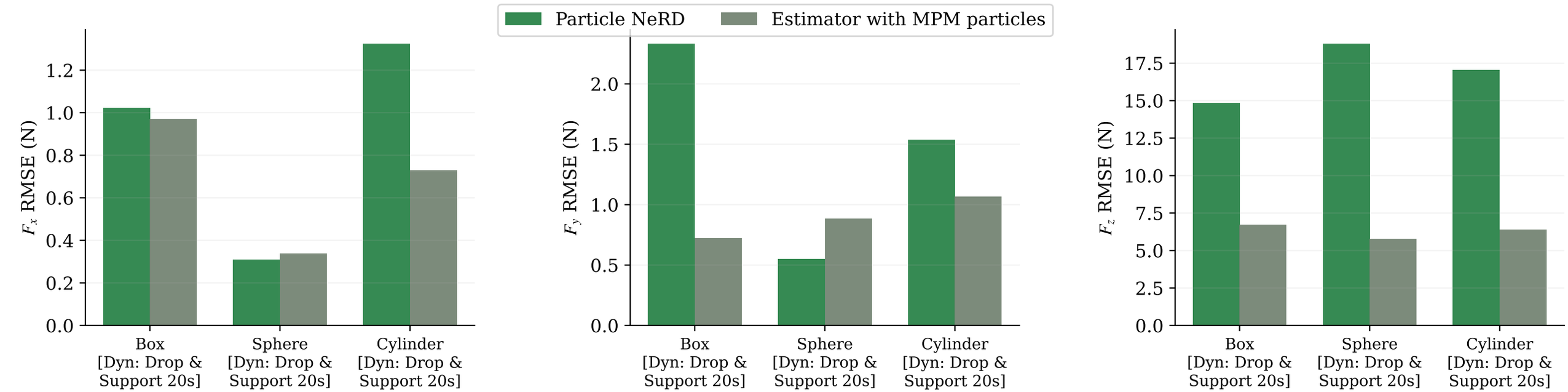

Figure 5: Frozen object-model checkpoint 6,000. Dynamic Motion: 2 kg object held until 1 s, then released to drop and settle on sand through 20 s. Columns show Fx, Fy, Fz. Top row: component RMSE; bottom row: signed mean error (estimate minus native MPM material force).